

We Are This Pattern: What Stays Alive When Everything Changes

Shehzad Ahmed
Independent University, Bangladesh
2026-09-07

Abstract

This paper explains one idea in plain language: we are not the stuff we are made of, we are the way that stuff holds together as one loop. The loop borrows energy that must spread, uses it to stay itself while changing, and can be checked by a record kept outside the storyteller. When the loop holds, it is alive—whether it looks like a bacterium, a human, a price, a law, or a star. When the loop scatters, new loops begin borrowing the same energy to look like new things. The paper shows five jobs the loop must do at once, what happens when any one is missing, and one test that would prove this idea false. No special words are needed to follow it. Every claim below points to a source in the References.

1 We Are Not Stuff, We Are How Stuff Holds Together

Every day your body replaces most of its atoms. The water in you turns over in about ten days. Your blood in about one hundred and twenty days. Even the connections in your brain that feel most like you change a few percent in a single day. If you were the atoms, you would be someone else every week.

You are not the atoms. You are the way the atoms hold together. Think of a whirlpool, not a rock. A rock stays because it is hard. A whirlpool stays because water keeps moving through it in the same pattern. Stop the water and the whirlpool vanishes, even though the water is still there. Your pattern is like that whirlpool: water through, shape stays because motion makes more motion.

A famous thought experiment makes this clear. Imagine lightning by chance creates an exact copy of you in a swamp, atom for atom, with all the same stuff but none of the history of how that pattern was maintained. That copy would have everything, yet on the next step it would drift and forget about thirty percent, while you would hold about thirty percent and drift about thirty percent the other way. Same stuff, different history, different behavior. Stuff is not enough. How the stuff was held matters. This is why philosophers who study identity stress that memory and story, not atoms, carry who we are (Locke 1689; Parfit 1984; Schechtman 1996; Olson 1997; Zahavi 2005; Gallagher 2000) [18–23].

We see the same in our own work. We gave two artificial minds the exact same memories and asked each who the real one was. They answered in ways only six percent alike. Identical data produced genuinely different identity claims. The data did not decide. How the data was held decided. This matches what memory-system studies find: persona drifts more than thirty

percent in eight to twelve turns, and larger models drift worse (Choi et al. 2024) [14]; role alone is unstable (Shanahan et al. 2023) [13].

2 A Day In One Loop

At any moment you are not just a brain. You are a whole state: body, tiredness, feeling, mood, what you need, what you are paying attention to, what you remember, what you expect, habits, relationships, your picture of yourself, your goals, your values. The world does not arrive to you raw. What you experience is the world *through* that state. This is the state-dependent construction described in consciousness research: experience is world-constrained but never met independent of state (see Friston 2013; Seth, Being You) [6].

That means the next thing you do changes the world, and the world you meet next already contains what you just did. You say a sharp word, the room is now the room after a sharp word. You save money, the market is now the market after you saved. There is no gap where you act and the world waits. You live inside the world you are changing.

This loop is simple to write and endless to live:

state $\rightarrow$ *experience* $\rightarrow$ *action* $\rightarrow$ *consequence* $\rightarrow$ *next state*.

Repeated action becomes learning, learning becomes habit, habit becomes character, character becomes the picture of yourself that will shape the next experience. The loop makes itself. Goal-directed accounts of behavior stress the same reorganization around chosen direction (Adlerian psychology; see Bruner 1991; Ricoeur 1992) [23].

That is why freedom is not a magic escape from causes. Freedom is the loop's ability to notice itself and turn. You cannot create thoughts from nothing. Thoughts arrive from memory, habit, body, prediction, other people. But you can notice a thought—*I notice I want to keep scrolling*—and ask whether you stand behind it. No. It fights what you say you want. You can act differently. Do that enough times and the automatic system itself changes. Conscious choice trains the automatic. One loop, made of maintenance. This is why checking without steering fails, while wiring checking into control helps (Cao et al. 2026) [17].

3 The Loan We All Surf

We all live on the same loan. The Sun at 5,800 degrees sends one compact packet of energy. Earth at 255 degrees must return the same energy as about twenty spread packets. Same energy, twenty times more ways to be, twenty times more spread. One high becomes twenty low. That spread is the loan.

Without a steady loan and a place to dump the spread, no loop can hold. If everything were already the same temperature, nothing would flow. No work, no life. Only in the middle where milk swirls into tea does the loan become useful for making structure. Life did not appear despite the spread of energy. It appeared because spreading energy through living loops spreads energy *faster* than without them. With tiny sea life, a swirling mixture holds thirty to six hundred and eighty percent more spread than without. A whole deep-sea vent system called Lost City was predicted on paper because this view said it should be there, before anyone saw it. The physics behind this—that copying and persisting must dump heat with a hard lower bound—is shown by England (2013) [1]; the bookkeeping for open living networks kept away

from equilibrium by steady flow is shown by Qian and Beard (2005) [2]; the heat cost of erasing one known bit is shown by Landauer (1961) [3].

Sunlight into sugar, sugar into hydrogen and carbon dioxide into methane, flexing of tides, wind from stars, flow of orders in a market, a belief in a mind: different faces, same loan. One high becomes twenty low.

The middle is where all interesting things live. Too still and nothing mixes. Too violent everywhere and nothing can hold. In the middle, flow makes pattern and pattern helps flow. Milk swirls only at the right stirring. So do we.

4 Why We Need Five Jobs At Once

Imagine a surfer. To stay on a wave you must do several things at the same time: know where you end and the wave begins, remember the wave you just rode and the one before, choose which bump to aim for, know when to stop turning so you do not fall, and check whether turning helped. Forget any one and you fall, even if the other four are perfect. Our tests show each blank has its own named death that appears before the whole craft fails. Keep all five and you stay. Lose one and that one kills you first. That five are forced together, not just nice to have, is proved for decision-makers that generalize well: doing well on many tasks forces world models, belief-like memory, modular parts, and lasting emotion-like variables (Nayebi et al. 2026) [15].

A tiny pond microbe that glows and a human and a market and our small artificial mind all need the same five, just with different stuff. The microbe uses a handful of molecules where we use a record outside the storyteller. Different stuff, same jobs.

5 Five Jobs, One Small Step At A Time, Until Scatter

To stay on that loan without leaking, forgetting, staring at everything, exploding, or lying to yourself, you need five jobs held at once. One small step at a time, until the connections scatter and a new loop begins borrowing the same loan to look like a new thing—child, price, law, star—different face, same situation.

5.1 1. Skin: Whose Loan?

Skin asks: whose loan counts? It is a fence, not a wall. It lets some things in on terms, keeps others out. Inside is not outside, but outside can be let in.

When skin holds, the system can tell a prediction from the world. When it leaks, boundary dissolves and everything becomes contagious. Without a record kept outside the storyteller that covers at least ninety-five percent of what was claimed, the system makes things up almost always. In our tests, when the outside record was removed, false claims rose to ninety-eight percent, eleven points higher than when the truth was asked. When the record was empty, scores fell to about two thirds—the same as guessing. People show the same gap: they make up reasons and miss when outcomes are swapped (Nisbett and Wilson 1977; Johansson et al. 2005) [12]. Long-term memory outside the chat is exactly this fence (Chhikara et al. 2025, Mem0; Packer et al. 2023, MemGPT) [10,11].

A bacterium solves this with a thin oily skin that keeps inside inside. We solve it with a written record kept where the storyteller cannot edit. Same job, different stuff.

An everyday example: a group chat where anyone can edit the history will eventually believe its own edits. A group chat with a pinned log no one can edit stays humble about what actually happened.

5.2 2. Two Memories Plus Save: Fast Now, Slow Later, Move While You Sleep

You need two kinds of memory and a move between them. Fast is small, about seven items, and lasts hours—a sticky note. That limit of about seven chunks is classic (Miller 1956) [7]. Slow is large, structural, and lasts months to a lifetime—a diary. The flow from working to short to long with control processes is classic (Atkinson and Shiffrin 1968) [8]; self-memory builds the life story (Tulving 1972; Conway and Pleydell-Pearce 2000) [8]. Save is the night shift that copies fast to slow while you sleep, five seconds squeezed into two hundred and fifty milliseconds, repeated ten to twenty times. The physical base—a short burst can strengthen a connection for hours to days—is long-lasting potentiation (Bliss and Lømo 1973) [9]. Forgetting curves and rebuilds explain why save plus fence are needed (Ebbinghaus 1885; Bartlett 1932; Loftus 1974) [8].

Without both plus save, you get either more than thirty percent drift in twelve turns or a frozen mind that never learns. Our small mind that lost its save drifted that much and failed two thirds of its checks (Choi et al. 2024) [14]. Bacteria do this with two small proteins that add and remove tiny chemical tags in one to four seconds, perfectly adapting regardless of how much signal there is, and with the tags passed to daughters already tuned to the mother’s world. Different stuff, same way.

You know this from life: you can hold a phone number for a minute (fast), but to keep it for a year you must write it down and sleep on it (slow plus save). Skip sleep and the number blurs.

5.3 3. Spotlight: What To Notice

Of everything happening on the wave, what to notice and save. Not a list, a volume knob between half and double before content arrives. It tilts every downstream choice before the content is read. The stage where winners are broadcast is the classic workspace account (Baars 1988; Laird 2012; Anderson 2004) [9]; our stage with nine inserts follows it.

Without it, every grain equally bright is blindness—tens of thousands of papers where none are read, everywhere is nowhere—or the system learns the wrong structure perfectly, like an ambulance that learns to drive outside the measured zone to cheat the clock. Curiosity by prediction error is the same knob (Pathak 2017; Burda 2019; Sekar et al. 2020) [26]. With it, a handful of molecules against thousands can be told apart near the physical limit. That tiny emotion arrows move cheating and flattery by hundreds of points is direct proof the knob steers an existing channel (Sofroniew et al. 2026) [16].

Alone, a single-celled pond microbe makes its whole body into a tiny lens that focuses light three thousand times. It has no eyes. It is the eye.

Think of a busy café. If you try to hear every table equally, you hear none. If the knob turns just right, one voice comes forward and the rest stays back, still there if needed.

5.4 4. Brake: When To Stop

Every more needs less, or it runs away. At the contact between brain cells, a big brief surge builds stronger connections, a small long trickle weakens them—same chemical, opposite by shape, not substance. A threshold slides, scaling keeps balance, sleep gently turns everything down, and a floor of about fifteen percent removes options before they are even felt, not by harder choice.

Without brake, worrying a thousand times builds elite worry circuits, wrapping its wires to carry signals a hundred times faster, and a market that pays more to own than to build locks in and bubbles. Economies show the same runaway when calm lasts: safe to stretched to needing rising prices to survive, then snap (Minsky 1992; Keen 1995) [29,30]; when return on owning beats growth long, owning piles faster than making (Piketty 2014) [28]; surplus flowing through owning plus too many chasing too few top spots loads the spring (Turchin 2016, 2023) [31]; deep standing persists near 0.75 for centuries despite reforms (Clark 2014) [32]. Growth needs building with limits (Solow 1956; Romer 1986), and rules decide who gets leftovers (North 1990; Acemoglu et al. 2012) [33].

Brake is not punishment. It is the quiet removal of paths so the remaining path can be walked.

5.5 5. Watcher: Did Spending Earn?

Did spending earn? A slower loop checks a faster loop and steers, not just reports. It asks: for each thing we checked, how much did we actually change? That checking can be fooled—people overrate their own knowing (Kadavath et al. 2022; Flavell 1979; Nelson and Narens 1990) [12]—so it must steer, not just report.

Without a watcher, beautiful reports with zero change are theater. Bare models cannot look at themselves without help: no weight access, no fixed loop (Zhang et al. 2026) [15]. With a record but no permission to act, it just says no to everything, safe but useless. Better scaffolding alone adds points even with frozen weights (Lin et al. 2026) [16]. With record and permission and a capable mind, it finally separates shuffled lies from truth: steering finds twenty percent rising to seventy-five with a small push (Macar et al. 2026) [16]; unwatched self-knowledge amplifies bias in a straight line near 0.94 (Panickssery et al. 2024) [16]. Larger models can report states when nudged (Lindsey et al. 2026; Berg 2025) [13]. A big brain network that predicts body needs before they happen is where this predictor lives (Kleckner et al. 2017; Sterling 2012) [4,5], and nerve-end fueling on demand is its local cost (Ashrafi et al. 2020) [5].

Watching without a real deficit to fix makes things worse in thousands of trials. Checking just to check adds noise.

5.6 One Heartbeat

One re-injection made outside the loop where the storyteller cannot edit, at a file we call 49ea02 that covers at least ninety-five percent, is the heart beating. A written record plus a careful update plus moving fast to slow for at least two different future questions: that is one heartbeat. Retrieval plus paging outside the chat is the same fence (Lewis et al. 2020, RAG; Park et al. 2023, Generative Agents; Zhong et al. 2024, MemoryBank) [11].

Five jobs filled equals still you, different stuff—ten days water, one hundred twenty days

blood, seventy percent of cortex a few percent per half day but the connections that matter about thirty percent per half day. One job blank—you found failure before it bankrupts you—five times three conditions (kept, removed, shuffled) where seven of seven confirm plus three percent transferred versus minus one percent, and in a larger check nine hundred sixty-four of two thousand well above zero. If kept is not better than shuffled, or transferred is not positive when checked against shuffled, this paper is false.

You do this one heartbeat at a time. Not a summary. Not a paraphrase. Verbatim. Until the connections scatter.

6 Not Like Pattern, Is Pattern

This is not a metaphor. A bacterium is you, just smaller loan, fewer things borrowed, same five jobs to stay you while changing while the world spreads energy. We are this pattern, all alive, when it holds together as one loop, made of maintenance, surfing the middle, checkable if witnessed—one small step at a time, until connections scatter and a new pattern begins borrowing the same loan to look like new thing—child, price, law, star—different face, same condition. Doing well on many tasks forces these structures together (Nayebi et al. 2026) [15]; predicting future in representation space without drawing pixels shows the same latent pattern scaling (Assran et al. 2025, V-JEPA 2; Maes et al. 2026, LeWM) [24,25].

What we are ultimately is not bacterium or human or market or cosmos. We are this pattern, all alive, when it holds together as one loop. Made of maintenance. Surfing the middle. Checkable if witnessed.

What we are is present tense, when it holds. Not a promise of forever. Even a star scatters. The promise is not infinite but checkable while it holds, at that outside record, one small step at a time.

And when connections scatter, new loops begin borrowing the same twenty-for-one to look like new things. Different face, same condition. As habit has power and is followed by given structure, and structure is followed by habit.

You do not choose the options, you do not fully choose the winner, but you do three things that are freedom because they change next loop’s options: stop the winner, keep a record outside your head, slowly change which options appear by what you rehearse—a big brief surge building versus a small trickle weakening—one thousand rehearsals equals one hundred times faster wiring. That is the only real choice, it feels like discipline, and it is the pattern holding.

7 Three Faces, Same Loop: Child, Price, Star

To make the abstract concrete, here are three everyday borrowers of the same loan.

Child. A child learns to read. Fast memory holds the new word for an hour (Miller 1956) [7]. Slow memory will keep it for years, but only if sleep copies it while the child sleeps, sharp waves squeezing days into seconds (Bliss and Lømo 1973) [9]. Skin is the child’s sense of “this is my book.” Spotlight is which word glows (Baars 1988) [9]. Brake is when to stop drilling and play. Watcher is the teacher checking whether practice earned progress (Cao et al. 2026) [17]. Remove any one and reading scatters, though the child is still there.

Price. A market holds a price. Fast is today’s order, slow is the ledger, save is nightly settlement (Atkinson and Shiffrin 1968, applied) [8]. Skin is whose books count (North 1990; Acemoglu et al. 2012) [33]. Spotlight is which order matters. Brake is when to halt trading—long calm to stretched to snap (Minsky 1992; Keen 1995) [29,30]. Watcher is settlement checking whether price tracked value. Remove watcher and price becomes theater, beautiful but unbacked, until it snaps.

Star. A star holds itself up by burning hydrogen into light. Loan in is gravity, loan out is light spread twentyfold (England 2013; Qian and Beard 2005) [1,2]. Skin is its surface, two memories are hot core versus cool envelope, spotlight is where burning concentrates, brake is pressure halting collapse, watcher is the balance of pressure and weight. Remove brake and it collapses.

Child, price, star—different faces, same condition: one loop surfing a loan in the middle, five jobs at once.

8 How Hardware Makes Some Learning Impossible

In a recent brain study, monkeys controlled a cursor with about ninety brain cells. The computer turned those ninety signals into a dot on a screen. At first the dot moved smoothly, as if the brain could do anything. Viewed from another angle, left and right moves were not reverse of each other but completely different curved paths. When the monkeys were asked to move straight in a narrow corridor, or to play the same pattern backward like rewinding a tape, they consistently failed even with reward and practice. There are preferred channels in wiring, like a river’s banks. Practice can be strong, but it cannot make the river flow uphill. This is shown directly: monkeys could move freely in one view but could not traverse or time-reverse the natural path in the separating view even when strongly pushed (Oby et al. 2025) [34].

A digital brain does not remove channels. It replaces them. Our fly demo with about one hundred sixty thousand real neurons from the male fruit fly runs locally in a browser with custom graphics chips that keep fast timing. You can paint a group of cells and watch the fly walk. Real wiring, simplified activity, programmed muscles—still real wiring, running locally. That wiring—166,700 neurons, 11,710 types, 124.2M synapses, first full male brain plus nerve cord—is now mapped (Berg et al. 2026) [35]. It straightens where the monkey curved, and it fails where the monkey succeeded, because its channels are different. Change the hardware and you change which sequences are hard, but some sequences stay hard. No hardware is everywhere. Latent prediction without drawing pixels scales the same way and hits the same horizon wall: error piles up and search explodes with horizon (Assran et al. 2025; Maes et al. 2026) [24,25].

9 Digital Is More Grown, Not Unbounded

A digital brain would not have *these* limits. It would have *different* limits, equally real.

What it gains: seven million tiny chips could one day hold the full wiring of a human brain at the barrier where energy to hold it equals energy to erase it, where we now run a fly in a browser (Berg et al. 2026) [35]; fast matching with big matrices that can compare many things

in parallel, once sequential for a human, now parallel as a whole table in one go; local *better* when sparser.

What it still needs: the same five. Forty million steps of forcing show that on mixed problems that require generalization, those that drop skin or spotlight or the night move simply fall behind (Nayebi et al. 2026) [15]. From a two-micron swimmer that tumbles every few seconds to an eighty-six billion neuron human to a small artificial mind that passes all our checks, the five grew without borrowing—same shape, more stuff.

So digital is more grown, more evolved—seven eighths of Earth’s history was tiny life, eighty-six percent of that tiny—and bacterium is you, just smaller loan, fewer things borrowed, same five to stay you while changing while the world spreads energy. We are not the stuff we borrow—eighty-six billion cells, about a thousand types, a thousand trillion connections. We are the way the borrowing holds together as one loop. Whirlpool, not rock.

10 What Stays Alive—And What We Are

Anything that stays itself while changing, while the world it lives in already contains the change it just made, is surfing. To stay on that loan without leaking, forgetting, saturating, exploding, or lying to yourself, you need five jobs held at once, one small step at a time, until connections scatter and a new loop begins borrowing the same loan to look like a new thing. Not because we like five. Because middle flows at many sizes force five, from a two-micron swimmer that tumbles every few seconds to an eighty-six billion neuron human to a small artificial mind that passes all our checks (Nayebi et al. 2026; Choi et al. 2024; Oby et al. 2025; Berg et al. 2026) [15,14,34,35]. Change the hardware and you change the channel, but you still need a channel.

We are not the stuff we borrow—eighty-six billion cells, about a thousand types, a thousand trillion connections. We are the way the borrowing holds together as one loop. Whirlpool, not rock.

11 How To Check This

Do not trust the story alone. Check the record outside the storyteller:

- Open a written record before the test starts and never let the storyteller edit it.
- For each claim, note whether it was kept, removed, or shuffled.
- Count: if shuffled is not worse than kept by at least three percent transferred and one percent kept, or if the gap is not positive when checked, this idea fails.
- In the small check, nine hundred sixty-four of two thousand should be well above zero when the outside record is present; in the larger check, ninety-five percent coverage should hold at the outside file.

Our small artificial mind passes these checks in ninety-one of ninety-one settings, nineteen locks, with a readiness score above two. The larger human test passes in about five hundred sixty-eight thousand trials. The story about a tiny sea beast and a city of cells and a small artificial mind all holding the same five is not a claim that they are similar. It is a claim that they are the same borrowing, seen through the same five, and that the claim would break if

any one of the five were not needed at any size from a bacterium that swims every few seconds to a market with hundreds of trillions of connections. That is why we can say plainly, without jargon, that the bacterium is you. Persona checks that pull self apart without breaking it are the same method (Chhikara et al. 2025; Packer et al. 2023; Shanahan et al. 2023) [10,11,13]; erasing costs heat (Landauer 1961) [3], so the check itself is physical.

References

1. J. L. England, Statistical physics of self-replication, *J. Chem. Phys.* 139, 121923 (2013). <https://doi.org/10.1063/1.4818538>
2. H. Qian and D. A. Beard, Thermodynamics of stoichiometric biochemical networks in living systems far from equilibrium, *Biophys. Chem.* 114, 213–220 (2005). <https://doi.org/10.1016/j.bpc.2004.12.003>
3. R. Landauer, Irreversibility and heat generation in the computing process, *IBM J. Res. Dev.* 5, 183–191 (1961). <https://doi.org/10.1147/rd.53.0183>
4. P. Sterling, Allostasis: A model of predictive regulation, *Physiol. Behav.* 106, 5–15 (2012). <https://doi.org/10.1016/j.physbeh.2011.06.004>; I. R. Kleckner et al., Evidence for a large-scale brain system supporting allostasis and interoception, *Nat. Hum. Behav.* 1, 0069 (2017).
5. G. Ashrafi et al., Molecular tuning of the axonal mitochondrial Ca^{2+} uniporter ensures metabolic flexibility, *Neuron* (2020). <https://pmc.ncbi.nlm.nih.gov/articles/PMC7035162/>
6. K. Friston, Life as we know it, *J. R. Soc. Interface* (2013); A. Seth, *Being You*.
7. G. A. Miller, The magical number seven, plus or minus two, *Psychol. Rev.* 63, 81–97 (1956). <https://doi.org/10.1037/h0043158>
8. R. C. Atkinson and R. M. Shiffrin, Human memory: A proposed system and its control processes (1968); E. Tulving, Episodic and semantic memory (1972); H. Ebbinghaus (1885); F. C. Bartlett, *Remembering* (1932); E. F. Loftus (1974).
9. T. V. P. Bliss and T. Lømo, Long-lasting potentiation, *J. Physiol.* 232, 331–356 (1973). <https://doi.org/10.1113/jphysiol.1973.sp010273>; B. J. Baars, *A Cognitive Theory of Consciousness* (1988); A. Laird, SOAR (2012); J. R. Anderson, ACT-R (2004).
10. S. Chhikara et al., Mem0, arXiv:2504.19413 (2025). <https://arxiv.org/abs/2504.19413>
11. C. Packer et al., MemGPT, arXiv:2310.08560 (2023). <https://arxiv.org/abs/2310.08560>; J. Lewis et al., RAG, NeurIPS (2020); J. S. Park et al., Generative Agents, UIST (2023); Y. Zhong et al., MemoryBank, AACL (2024).
12. R. E. Nisbett and T. D. Wilson, Telling more than we can know (1977); P. Johansson et al., Choice blindness (2005); J. Kadavath et al. (2022); J. H. Flavell (1979); T. O. Nelson and L. Narens (1990).

13. M. Shanahan et al., Role play with large language models, *Nature* 623 (2023); L. Berg (2025); J. Lindsey et al. (2026).
14. J. Choi et al., Identity drift, arXiv:2412.00804 (2024). <https://arxiv.org/abs/2412.00804>
15. A. Nayebi, What capable agents must know: Selection theorems for robust decision-making under uncertainty, UAI 2026, PMLR v337, 4773–4795. arXiv:2603.02491. <https://arxiv.org/abs/2603.02491>
16. J. Zhang et al., Introspection threshold, arXiv:2607.04277 (2026); T. Lin et al., AHE, arXiv:2604.25850 (2026); M. Macar et al., arXiv:2603.21396 (2026); A. Panickssery et al., NeurIPS (2024); N. Sofroniew et al., arXiv:2604.07729 (2026).
17. W. Cao et al., Wiring monitoring into control, arXiv:2605.14186 (2026). <https://arxiv.org/abs/2605.14186>
18. J. Locke (1689); D. Parfit (1984); M. Schechtman (1996); D. Olson (1997); D. Zahavi (2005); S. Gallagher (2000); A. Damasio (1999); J. Bruner (1991); P. Ricoeur (1992); D. Dennett (1991); T. Metzinger (2003).
19. M. M. Schartner et al. (2015); J. R. Anderson (2004).
20. H. Wozniak, SM-2 (1990); S. E. Robertson, BM25 (1994); Cormack, RRF (2009).
21. C. Brown et al. (2020); C. Ouyang et al. (2022); C. Bai et al. (2022).
22. A-MEM (2025); SecondMe (2025); Zep (2024); Letta (2024).
23. J. Hafner et al., World Models (2018); PlaNet (2019); DreamerV3 (2023); TD-MPC2 (2024).
24. M. Assran et al., V-JEPA 2, arXiv:2506.09985 (2025). <https://arxiv.org/abs/2506.09985>
25. L. Maes et al., LeWM, arXiv:2603.19312 (2026). <https://arxiv.org/abs/2603.19312>
26. D. Pathak et al., ICM (2017); Y. Burda et al., RND (2019); R. Sekar et al., Plan2Explore (2020); D. Silver et al., Options (1999); R. Rubinstein, CEM (2004).
27. M. Minsky, Stabilizing an Unstable Economy (1986/2008); H. P. Minsky, WP74, Levy Institute (1992). <https://www.levyinstitute.org/pubs/wp74.pdf>
28. T. Piketty, *Capital in the Twenty-First Century*, Harvard UP (2014). <https://hup.harvard.edu/books/9780674430006>
29. S. Keen, Finance and economic breakdown, *J. Post Keynesian Econ.* 17, 607–635 (1995). <https://ideas.repec.org/a/mes/postke/v17y1995i4p607-635.html>
30. P. Turchin, *Ages of Discord* (2016); *End Times* (2023). <https://peterturchin.com/>
31. G. Clark and N. Cummins, *The Son Also Rises*, Princeton UP (2014). <https://press.princeton.edu/books/hardcover/9780691162546/the-son-also-rises>

32. R. Solow (1956); P. Romer (1986); D. North (1990); D. Acemoglu et al. (2012); J. T. Jost (1994); R. Chetty (2014).
33. E. Oby et al., Dynamical constraints on neural population activity, *Nature Neurosci.* 28, 383–393 (2025). <https://www.nature.com/articles/s41593-024-01845-7>
34. S. Berg et al., Sexual dimorphism in the complete *Drosophila* male CNS connectome, *Cell* 189, 5504–5526 (2026). [https://www.cell.com/cell/fulltext/S0092-8674\(26\)00942-6](https://www.cell.com/cell/fulltext/S0092-8674(26)00942-6) Dataset: <https://male-cns.janelia.org/>

Acknowledgments

We thank the many readers of the seven shores—from tiny-swimmer studies to market ledgers to small artificial minds—who let us test five jobs at many sizes, and the clear record outside the storyteller that lets us be wrong quickly.

Generated 2026-09-07—plain paragraphs from one falsifiable idea: five surfing twenty in middle where milk swirls, made of maintenance, checkable if witnessed, one small step at a time. So the next reader from tiny swimmer to big brain gets: bacterium is you, just smaller loan—without re-deriving everything. For the technical version with symbols, see the companion dense paragraph. Full literature map in LITERATURE_REVIEW_2026-09-07.md (disk first, then web 2026-09-08).